

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.4: Area of Polygons — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.4: Area of Polygons
Driving Question	DRIVING QUESTION: How can we calculate the exact area of any polygon-shaped surface — even when we cannot measure its height — so that builders, farmers, and designers in our community never order too much or too little material?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring the Courtyard Puzzle — Discovering Why "Base × Height ÷ 2" Is Not Enough	Learners observed a scale drawing and photographs of the Kericho community hall courtyard, noting that each triangular section has two labelled side lengths and one included angle but no perpendicular height. They observed the contractor's dilemma: the familiar area formula $A = \frac{1}{2} \times \text{base} \times \text{height}$ cannot be applied directly because no height is physically measurable.	Learners learned that the perpendicular height of a triangle can be expressed as $h = b \sin C$ when side b and the included angle C are known (h is the side opposite C in a right-angled sub-triangle whose hypotenuse is b), and that substituting this into the standard formula gives the sine-area formula $A = \frac{1}{2}ab \sin C$. They applied this formula to calculate the areas of Triangle A ($\approx 39.9 \text{ m}^2$) and Triangle B ($\approx 21.8 \text{ m}^2$) of the Kericho courtyard.	The sine-area formula explains how the contractor can calculate the area of each triangular section of the Kericho courtyard accurately without dropping a physical perpendicular — resolving the first part of the unit puzzle. The remaining courtyard shapes (parallelogram, hexagonal border) still need their own area formulas, which will be developed in the lessons ahead.
Lesson 2 Lesson 2: Triangle Area Without a Height — $\frac{1}{2}ab \sin C$ in Practice & Heron's Formula Introduced	Learners measured two sides and the included angle of triangular classroom objects (set-squares, diagonally-cut book covers, Kericho courtyard scale drawings) and computed areas using $\frac{1}{2}ab \sin C$, comparing results with rectangle-halving estimates. They also read the Rift Valley maize farm sub-phenomenon: three side lengths known, no angle, no height accessible.	Learners derived Heron's Formula [Area = $\sqrt{s(s-a)(s-b)(s-c)}$], where $s = (a+b+c)/2$] from the sine-area formula through a structured teacher-guided derivation, then applied it to Mwangi's triangular maize plot near Eldoret (sides 40 m, 55 m, 63 m → Area $\approx 1,087.7 \text{ m}^2$) and a second practice plot.	Learners updated their cumulative Kericho courtyard scale drawings by tagging each triangular section with the appropriate formula ($\frac{1}{2}ab \sin C$ or Heron's), recording computed areas, and writing a sentence explaining how many sections the contractor can now calculate without measuring a perpendicular height. The DQB was updated to show that area can be found with either (i) two sides and an included angle, or (ii) three sides alone.
Lesson 3 Deep Practice & Model	Learners measured three sides of real triangular objects in the classroom	Learners consolidated Heron's Formula — Area = $\sqrt{s(s-a)(s-b)(s-c)}$ where $s =$	Learners revised their cumulative Kericho courtyard area models by annotating each

Building: Heron's Formula Consolidated	environment (tripod stand tops, triangular tiles, roof outlines visible from windows) using rulers and tape measures, and recorded all measurements on a Group Measurement Sheet. They posted results on a shared class table on the chalkboard and conducted a gallery walk comparing computed areas across groups.	$(a+b+c)/2$ — and proved through proof-by-example that the formula is symmetric: the computed area does not change when the labels a, b, and c are swapped. They connected this to the Kericho courtyard context: the contractor can measure the three sides of each triangular section in any convenient order and still obtain the correct area without needing a perpendicular height.	triangular section with Heron's Formula, recording a computed area estimate, shading solved zones, and marking the parallelogram stage and hexagonal border as PENDING for future lessons. They updated the Driving Question Board with new evidence under WHAT WE NOW KNOW and posted specific questions about remaining unresolved zones under WHAT WE STILL WONDER.
Lesson 4 Area of Rhombus and Parallelogram — Measuring the Stage Floor Without Dropping a Perpendicular	Learners physically cut rhombus-shaped card offcuts into four right triangles and parallelogram-shaped card offcuts into two congruent triangles, measured diagonals and side lengths, and recorded results on a class poster — demonstrating that both shapes can be decomposed into triangles whose dimensions are directly measurable.	Learners derived and applied $A = \frac{1}{2}d_1d_2$ for the rhombus (from the perpendicular-bisecting diagonal property) and $A = ab \sin \theta$ for the parallelogram (as twice the area of one constituent triangle), and explicitly connected the parallelogram formula to the triangle formula $A = \frac{1}{2}ab \sin C$ established in Lesson 1.	Learners explained — in writing on their courtyard models and verbally during cold-calling — that the Kericho courtyard contractor can calculate the area of the parallelogram stage floor ($\approx 21.75 \text{ m}^2$) and rhombus fabric patches using only side lengths, diagonal lengths, and an included angle, without needing to physically measure a perpendicular height, because the sine of the included angle mathematically encodes the height.
Lesson 5 Practise & Apply — Rhombus and Parallelogram in Context	Learners examined two diagrams of the Kericho courtyard parallelogram stage — one with an included angle and no height, one with a height and no angle — and predicted which formula applied to each. They also worked through a structured three-part problem set involving the courtyard stage ($6 \text{ m} \times 4 \text{ m}$, 65°), a rhombus kanga patch from Nairobi's Gikomba market (diagonals 24 cm and 18 cm; also sides 15.23 cm and angle 56°), and a parallelogram school garden in Kisumu ($6.5 \text{ m} \times 4 \text{ m}$, 110°), gathering numerical evidence that both formulas yield consistent results when both data sets are available.	Learners learned that: (i) the sine area formula — $\text{Area} = ab \sin \theta$ — is the practical choice when perpendicular height cannot be physically measured, as is the case for the Kericho courtyard stage; (ii) the diagonal formula — $\text{Area} = \frac{1}{2} d_1d_2$ — is efficient when both diagonals are accessible (e.g. on a flat fabric or already-laid tile); (iii) sin of an obtuse angle remains positive, so the formula works for angles greater than 90° ; (iv) the two formulas are consistent — they yield the same area for the same shape when both data sets are correctly used; (v) the contractor needs to round tile counts up to avoid under-ordering material.	Learners explained through peer assessment, whole-class discussion, DQB annotation, and model-building why the Kericho contractor must use the sine formula for the parallelogram stage: the courtyard surface has no accessible perpendicular height, but two side lengths and the included angle can be measured with a tape and protractor. They articulated a decision rule — use the diagonal method when diagonals are measurable; use the sine method when sides and an included angle are known — and applied it to judge which formula is more practical in each real-life Kenyan context. They updated the cumulative courtyard model with the stage area ($\approx 21.75 \text{ m}^2$) and a tile-count estimate (88 tiles), advancing the class closer to fully answering the driving question.
Lesson 6 Area of a Regular Pentagon: Deriving and Applying $A = (5s^2)/(4 \tan 36^\circ)$	Learners observed a regular pentagon (the njugu tray sub-phenomenon) printed on 1 cm^2 grid paper and counted squares to estimate its area. They measured all five interior angles with a protractor, confirmed	Learners derived that the apothem (height) of each isosceles triangle equals $(s/2) / \tan 36^\circ$, because the apothem bisects the apex angle, forming a right-angled triangle whose base angle is 36° . Substituting into $A = (1/2)$	Learners explained, in speech-bubble sentences on their Courtyard Model, that the vendor can find the exact area of a regular pentagonal tray using only its side length s, because dropping a perpendicular

	the angle sum is 540° and each angle is 108° , and observed a projected diagram of the pentagon divided into 5 congruent isosceles triangles from the centre, identifying the apex angle as 72° and the base angles as 54° .	$\times \text{base} \times \text{height}$ for one triangle and multiplying by 5 yields $A = 5s^2/(4 \tan 36^\circ)$. For $s = 18 \text{ cm}$, $A \approx 557.3 \text{ cm}^2$; for $s = 9 \text{ cm}$, $A \approx 139.4 \text{ cm}^2$. $\tan 36^\circ$ appears specifically because the right-angled half-triangle has a base angle of 36° — half of the 72° apex angle.	from the centre to the midpoint of any side creates a right-angled triangle in which the base is $s/2$ and the base angle is 36° , making the apothem equal to $(s/2)/\tan 36^\circ$. The full pentagon area then follows by calculating the area of one of the 5 congruent triangles and multiplying by 5. The DQB was updated with the class consensus answer to 'Why does $\tan 36^\circ$ appear in the pentagon formula?'
Lesson 7 Area of a Regular Hexagon — Deriving the Formula from Equilateral Triangles	Learners traced hexagonal pencil cross-sections and honeycomb cell diagrams, subdivided them into 6 equilateral triangles, measured sides and angles to confirm congruence, and recorded data in a structured table — providing direct physical evidence that every regular hexagon is composed of 6 identical equilateral triangles.	A regular hexagon of side length s has area $A = (3\sqrt{3}/2)s^2$ — derived by multiplying the equilateral triangle formula $A = (\sqrt{3}/4)s^2$ by 6. This formula requires only the side length and no perpendicular height, making it directly applicable to the Kericho courtyard hexagonal border tiles where no height can be physically measured.	Learners explained why the hexagon formula is a direct descendant of the sine-area triangle formula from Lesson 1 ($A = \frac{1}{2}ab \sin C$, specialised to equilateral triangles), constructed a Formula Family Tree linking all five polygon area formulas developed across the unit, applied the hexagon formula to calculate tile area and adhesive box quantities for the Kericho courtyard, and updated the DQB with final answers — completing the full evidence-gathering arc of the unit.
Lesson 8 Synthesis, Model Building & Final Assessment: The Complete Kericho Courtyard	Learners examined the complete Kericho community-hall courtyard scale drawing — a floor plan divided into triangular sections (Triangle A, B, C), a parallelogram stage, and a hexagonal tile border — and used all given measurements (side lengths, included angles, and tile side length) to compute the area of every section.	Learners synthesised all five area-of-polygon formulas from the unit — (a) Area = $\frac{1}{2}ab \sin C$ for triangles with two sides and an included angle; (b) Heron's Formula for triangles with three sides; (c) Area = $ab \sin \theta$ for the parallelogram; (d) Area = $5s^2/(4 \tan 36^\circ)$ for the regular pentagon (a standalone formula from Lesson 6, not part of the courtyard layout); (e) Area = $(3\sqrt{3}/2)s^2$ for regular hexagonal tiles — selecting the correct formula for each section and computing a total courtyard area and contractor material cost in Kenyan shillings.	Learners explained that the Kericho contractor can calculate the exact area of every polygon section — and therefore order precisely the right amount of tiling material — without ever measuring a perpendicular height, because mathematical formulas (sine-area, Heron's, parallelogram, hexagon) allow area to be determined from whichever measurements are physically accessible: side lengths and angles alone are sufficient.

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